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7 A plane takes 2 hours 24 minutes to fly from London to Malaga.
The plane flies a distance of 1644 km.

(a) Work out the average speed of the plane.

24 minutes = 24 ÷ 60 = 0.4 hours

1 mark
Convert to hours.

Total time = 2.4 hours

Speed = distance ÷ time = 1644 ÷ 2.4

1 mark
Divide kilometres by hours.

1 mark
Average speed = 685 km/h.

685 km/h

(3)

(b) Show that a speed of 25 m/s is greater than a speed of 80 km/h.

25 m/s = 25 × 3600 m per hour

= 90 000 m per hour

= 90 km/h

1 mark
Multiply by 3600,
then divide by 1000.

90 km/h > 80 km/h, as required.

1 mark
Compare the speeds in the same units.

(2)

(Total for Question 7 is 5 marks)